KRSH SOLUTIONS

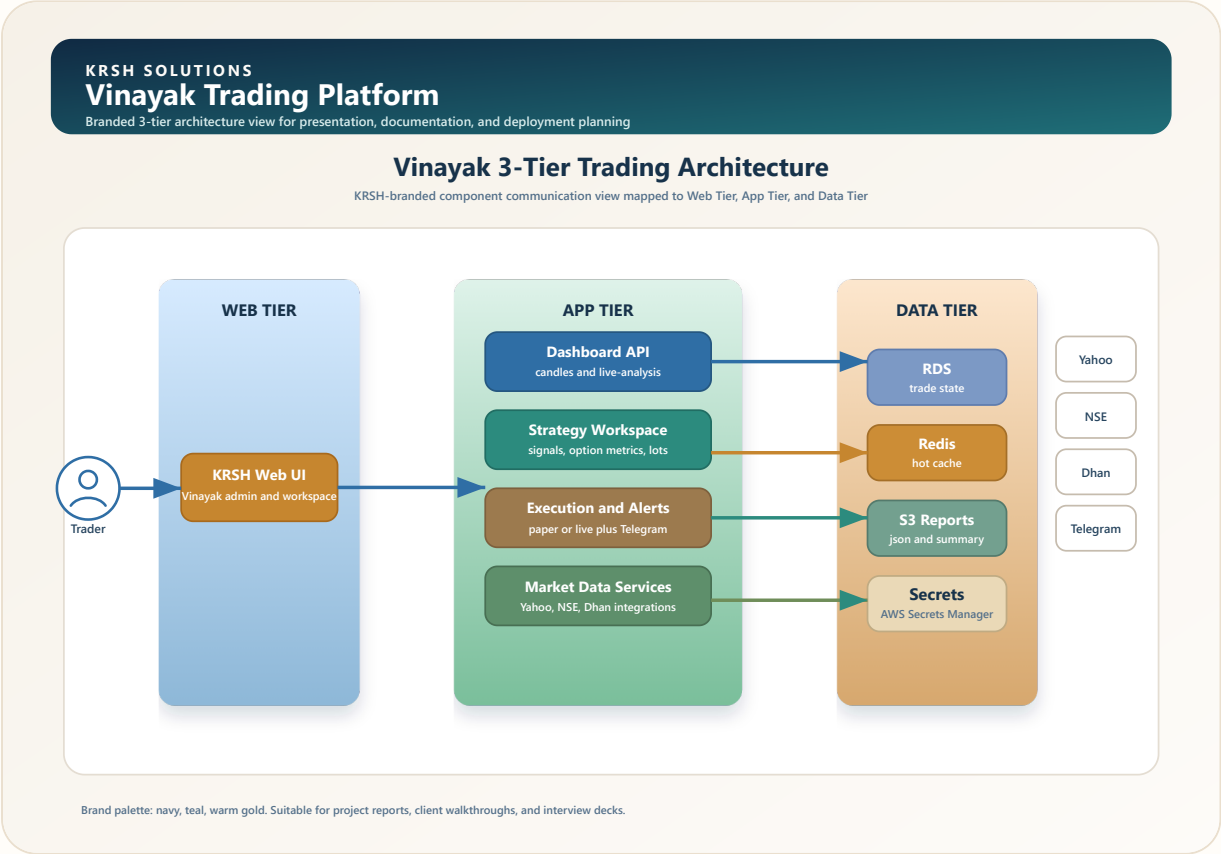
# Vinayak 3-Tier Project Guide

Branded documentation pack for the Vinayak trading platform. This version is formatted for project presentation, implementation planning, and architecture discussion.

Architecture | Implementation | AWS Ready | Trading Platform

## Architecture View

Vinayak now maps cleanly to a 3-tier structure. User traffic enters through the web layer, strategy and execution logic run in the app layer, and persistence plus cache plus report storage live in the data layer.




### Web Tier

- Public browser entry point

### App Tier

- FastAPI routes and services

- Admin and workspace pages
- Nginx or ALB reverse proxy

- Strategy workflow and execution
- Telegram, options, live OHLCV

## Data Tier

- RDS or relational DB layer
- Redis hot-cache
- S3 or local report artifacts

## Production Target

- ALB plus ECS Fargate
- RDS plus ElastiCache plus S3
- Secrets Manager and CloudWatch

> Current status: the project structure satisfies 3-tier logic. The remaining work is production environment verification and deployment hardening.

# Implementation Steps

## Step 1: Web Tier

Set up Nginx or ALB and route browser traffic to the Vinayak application.

```
sudo apt update
sudo apt install nginx -y
sudo systemctl enable nginx
sudo systemctl start nginx
```

## Step 2: App Tier

Install dependencies and run the FastAPI service behind the web layer.

```
py -3 -m pip install -r vinayak/requirements.txt
py -3 -m uvicorn vinayak.api.main:app --host 0.0.0.0 --port 8000
```

## Step 3: Data Tier

Configure database, Redis cache, and S3-backed report artifacts.

```
REDIS_URL=redis://localhost:6379/0
REPORTS_DIR=vinayak/data/reports
REPORTS_S3_BUCKET=your-report-bucket
AWS_REGION=ap-south-1
```

## Step 4: Final Validation

Validate browser flow, private app routing, data-tier connectivity, Redis cache population, and report artifact generation.